

1. Design an Airline Dynamic Pricing Engine

Problem Statement

An airline calculates ticket prices using the following rules:

Base Price = ₹5000

Modifiers:

- Business Seat → +40%
- Premium Economy → +20%
- Economy → No Increase

Booking Days:

- More than 30 days before travel → 10% discount
- Less than 7 days → 25% increase

Season:

- Festival Season → 20% increase

Age:

- Senior Citizen (60+) → 15% discount

Calculate the final ticket price.

Input Format

seat_type

booking_days

festival

age

Output Format

Final Ticket Price

Constraints

$1 \leq \text{booking_days} \leq 365$

$1 \leq \text{age} \leq 100$

Test Case 1

Input

Business

40

True

65

Output

6426.0

Test Case 2

Input

Economy

5

False

25

Output

6250.0

2. Create an Insurance Premium Calculator Based on Age, Health Score, and Vehicle Type

Problem Statement

An insurance company calculates annual premiums based on the following rules:

Base Premium = ₹10,000

Age Factor:

- Age below 25 → 20% increase
- Age between 25 and 50 → No change
- Age above 50 → 15% increase

Health Score:

- Health Score ≥ 80 → 10% discount
- Health Score between 60 and 79 → No change
- Health Score below 60 → 20% increase

Vehicle Type:

- Sports Car → 30% increase
- SUV → 15% increase
- Sedan → No change

Calculate the final insurance premium.

Input Format

age

health_score

vehicle_type

Output Format

Final Premium Amount

Constraints

$18 \leq \text{age} \leq 80$

$0 \leq \text{health_score} \leq 100$

Test Case 1

Input

22

85

Sports

Output

14040.0

Test Case 2

Input

55

55

SUV

Output

15870.0